

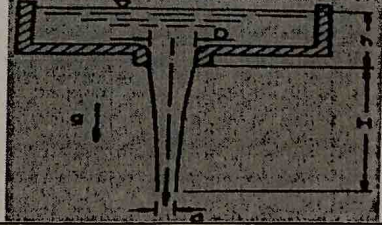
School of Engineering
Cochin University of Science And Technology

B.Tech Degree 3rd semester 1st series test in Mechanical Engineering Jan 2023
Time: 2 Hrs Sub: ME19-205 - 0304 Fluid Mechanics Max.Marks: 50
Faculty: Mr.Joshy P J & Dr. Gireesh K Thampi

CO- Course Outcome, PO -Program Outcome, L- Blooms Taxonomy level

CO1- Apply the knowledge of interfacial dynamics, capillarity, pressure and viscosity measurement.

CO2- Apply the knowledge of flow field lines (path lines, stream lines etc), Circulation & Vorticity and Bernouli's equation.

1(a)	Two large plane surfaces are 2.4 cm apart. The space between the surfaces is filled with glycerine. What force is required to drag a very thin plate of surface area 0.5 square metre between the two large surfaces at a speed of 0.6 m/s, if: (i) the thin plate is in the middle of the two plane surfaces, and (ii) the thin plate is at a distance of 0.8 cm from one of the plane surfaces? Take the dynamic viscosity of glycerine $= 8.1 \times 10^{-1} \text{ Ns/m}^2$. (8)	CO1	PO1 PO2	L2
1(b)	A pipe contains an oil of specific gravity 0.8. A differential manometer connected at the two points A and B of the pipe shows a difference in mercury level as 20 cm. Find the difference of pressure at the two points. (4)	CO1	PO1 PO2	L2
2(a)	The pressure outside the droplet of water of diameter 0.04 mm is 10.32 N/cm ² (atmospheric pressure). Calculate the pressure within the droplet if surface tension is given as 0.0725 N/m of water. (5)	CO1	PO1 PO2	L1 L2
2(b)	The dynamic viscosity of an oil, used for lubrication between a shaft and sleeve is 6 poises. The shaft is of diameter 0.4 m and rotates at 190 rpm. Calculate the power lost in the bearing for a sleeve length of 90 mm. The thickness of the oil film is 1.5mm. (8)	CO1	PO1 PO2	L1 L2
3(a)	Determine the equation of a stream line and equipotential lines passing through the point (a,0) if the two-dimensional flow is described by $u = \frac{-y}{b^2}$, $v = \frac{x}{a^2}$. Also plot streamlines qualitatively. [CO2]. (5)	CO2	PO1 PO2	L1 L2
3(b)	Consider a fluid flow through a well-rounded orifice positioned at the bottom of a large tank located at a height. Plot the stream lines and potential lines of the flow. [CO2]. (3)	CO2	PO1 PO2	L2
3(c)	Given the velocity field $u = u_0 \cos \omega t$, $v = v_0 \sin \omega t$ with $\frac{u_0}{\omega} = \frac{v_0}{\omega} = 1 \text{ m}$. Determine (i) stream lines for $\omega t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, (ii) the path lines (iii) the path line of a particle, which at time $t = 0$ is in the point $x = 0, y = 1 \text{ m}$. [CO2]. (5)	CO3	PO1 PO2	L3
4(a)	A cyclone is assumed to have the following velocity distribution: $v_\theta(r) = \begin{cases} \frac{\omega r}{r_0} & r \leq r_0 \\ \frac{r_0^2}{r} & r > r_0 \end{cases}$, $v_r = 0$. Given $r_0 = 10 \text{ m}$ and $\omega = 10 \text{ sec}^{-1}$. (i) sketch $v_\theta(r)$ Vs r , (ii) determine the circulation for a circle around the axis of the cyclone for $r < r_0$, $r = r_0$, $r > r_0$, (iii) show that for $r > r_0$ the flow is irrotational. [CO2]. (5)	CO2	PO1 PO2	L3
4(b)	Water flows out of a large reservoir under the influence of gravity in to the open air. Given $h = 0.1 \text{ m}$, $H = 1.5 \text{ m}$, $D = 0.1 \text{ m}$. What is the diameter 'd' of the water stream at the position 'H' below the opening. [CO2]. (4)	CO2	PO1 PO2	L2 L3
				
4(c)	A stream of water flowing at high speed from a garden hose. Explain what happened to the stream of fluid when the hose (i) held vertically up (ii) held vertically down. Also justify your answer. [CO2]. (3)	CO2	PO1 PO2	L2 L3

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